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Business Continuity & Disaster Recovery — Windows NLB Cluster, Failover & Recovery

Executive Summary

When one server dies, the business must not.

Module 17 turns high availability from theory into a working system. In a focused Windows lab, two identical IIS servers (Web Server 192.168.100.16 and Backup Server 192.168.100.17) are transformed into a resilient Network Load Balancing (NLB) cluster that keeps the website luxurytreats.com online no matter which node fails.

Students install the NLB feature on both machines, configure DNS suffixes and firewall rules, create a two-node cluster with a shared Virtual IP, and add the backup node. Name resolution is set via the hosts file on the client machine. The critical tests follow: normal operation is verified, the primary server is deliberately shut down, and the website continues serving without interruption. Finally, the failed node is brought back online to confirm seamless recovery.

Result: complete, hands-on proof of business continuity and disaster recovery using built-in Windows NLB — the exact technique that keeps critical web services alive during real outages.

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Implementing Business Continuity and Disaster Recovery Using NLB

Configure a two-node Windows Network Load Balancing (NLB) cluster so that the website luxurytreats.com remains available even if one server fails. This provides high availability and demonstrates a practical business continuity / disaster recovery technique.

Objective

Install Network Load Balancing features on both servers

Create a two-node NLB cluster with a shared Virtual IP

Configure name resolution using host's file

Verify continuous website access when the primary server is shut down

Confirm automatic failover and recovery

Machines Required

Machine	IP Address	Role
Web Server	192.168.100.16	Primary NLB node + hosts the website
Backup Server	192.168.100.17	Secondary NLB node + hosts the website
Attacker Machine	(any IP)	Client Test (browser)

Tools / Features Required

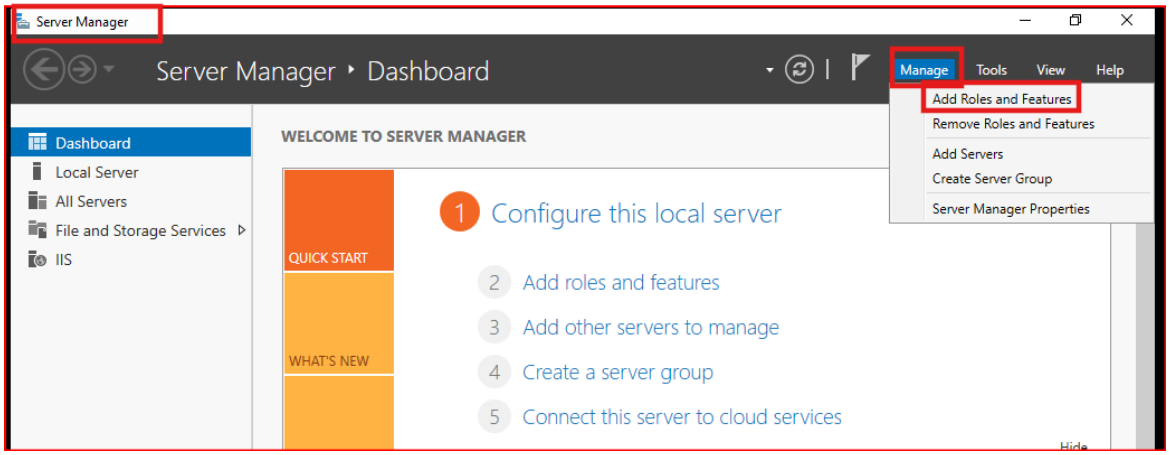
- Windows Server with IIS (website already working on both servers)
- Network Load Balancing feature (built-in)
- Hosts file for name resolution (no DNS server needed)

Important Prerequisites

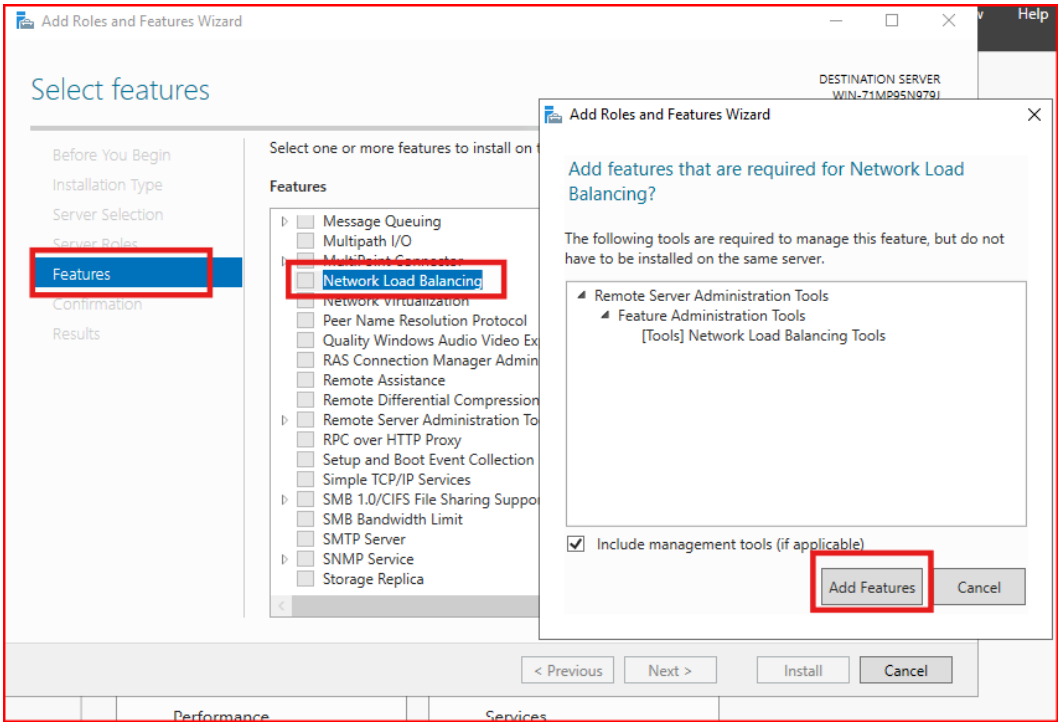
1. Both **Web Server** and **Backup Server** must:
    - Have the **same website** content for luxurytreats.com
    - Be able to ping each other
    - Have static IPs (192.168.100.16 & 192.168.100.17)
    - Have Administrator password set to Admin123
2. Preferred DNS on both servers can point to any working DNS (or leave blank for now).
3. Windows Firewall can be temporarily turned off on both servers for easier testing (recommended during lab).

Install Network Load Balancing Feature (Web Server)

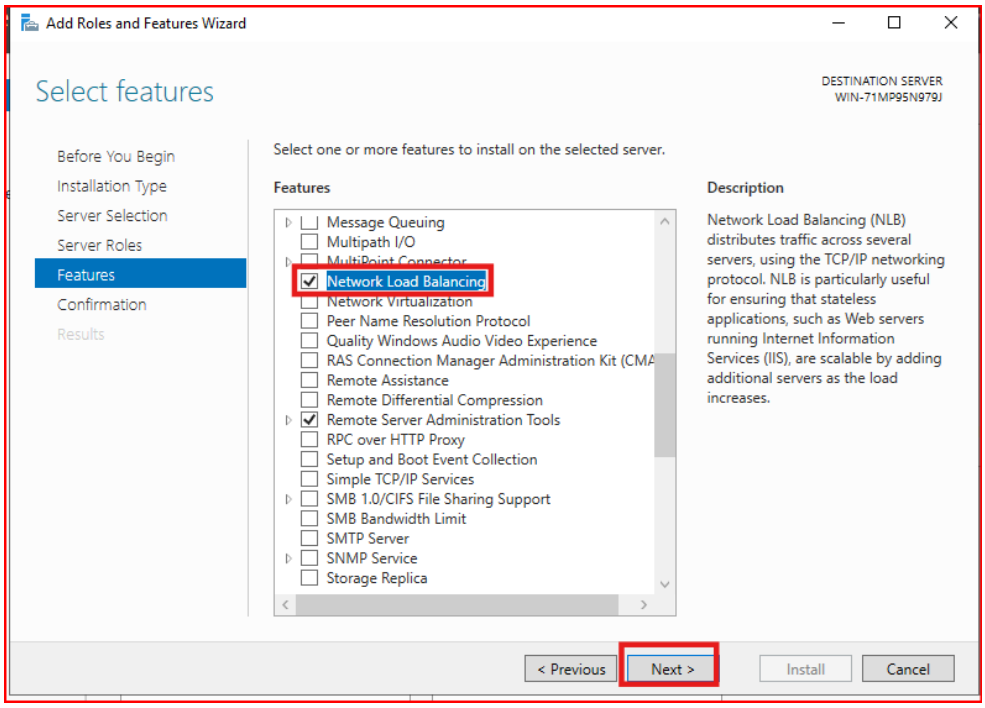
1. Log in to **Web Server** as Administrator / Admin123
2. Open **Server Manager**.
3. Click **Manage → Add Roles and Features**.



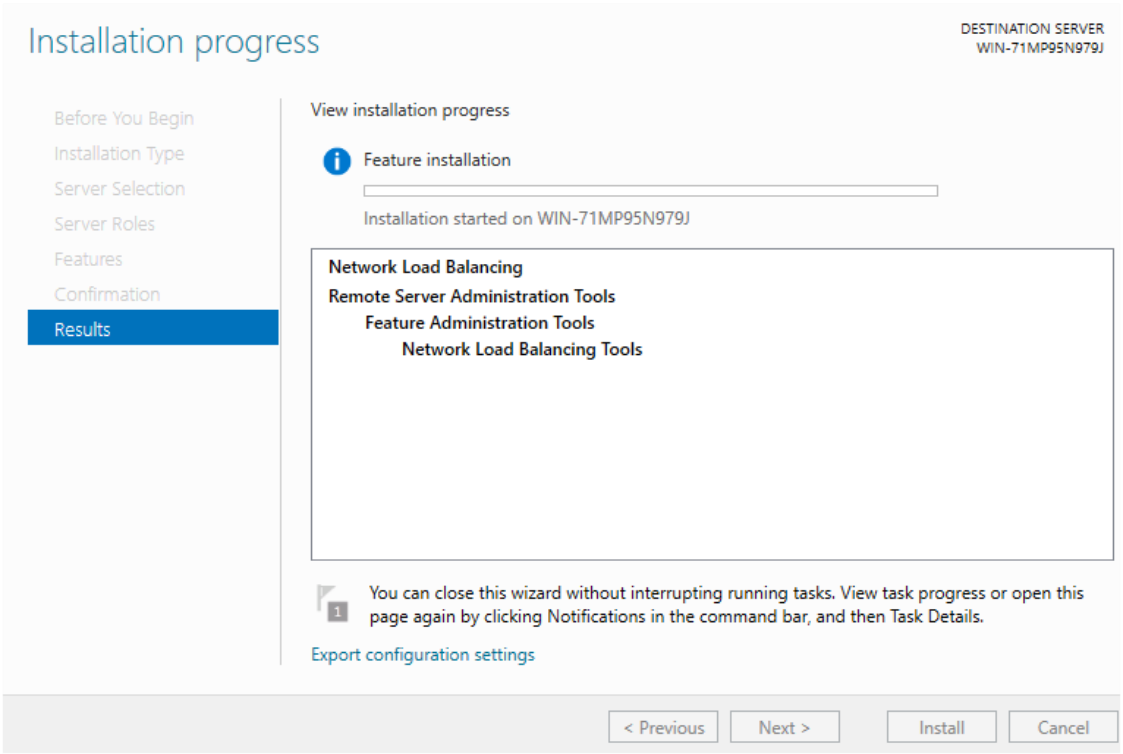
4. Click **Next** until you reach the **Features** page.
5. Scroll down and tick **Network Load Balancing**.
6. Click **Add Features** when prompted.



7. Click **Next → Install**.

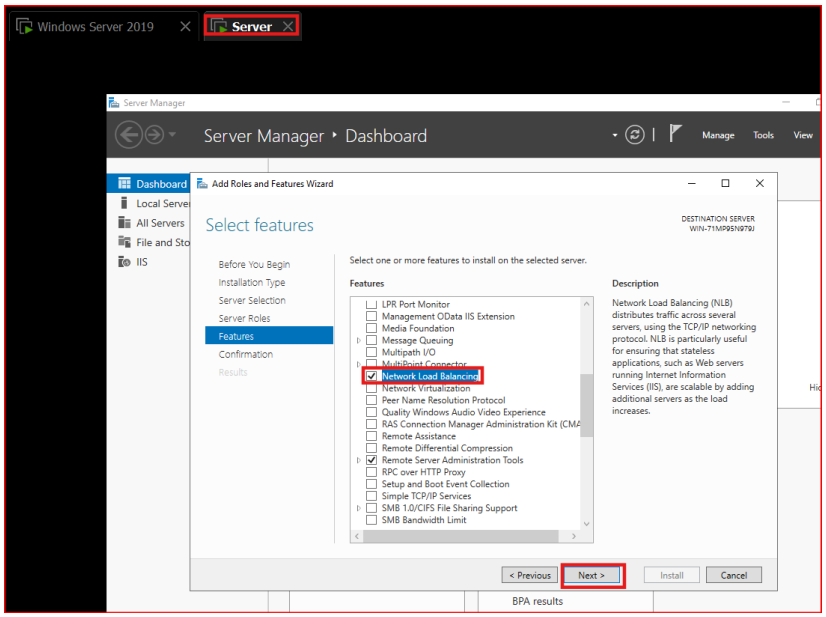


8. Wait until installation finishes → click **Close**.



Install Network Load Balancing Feature (Backup Server)

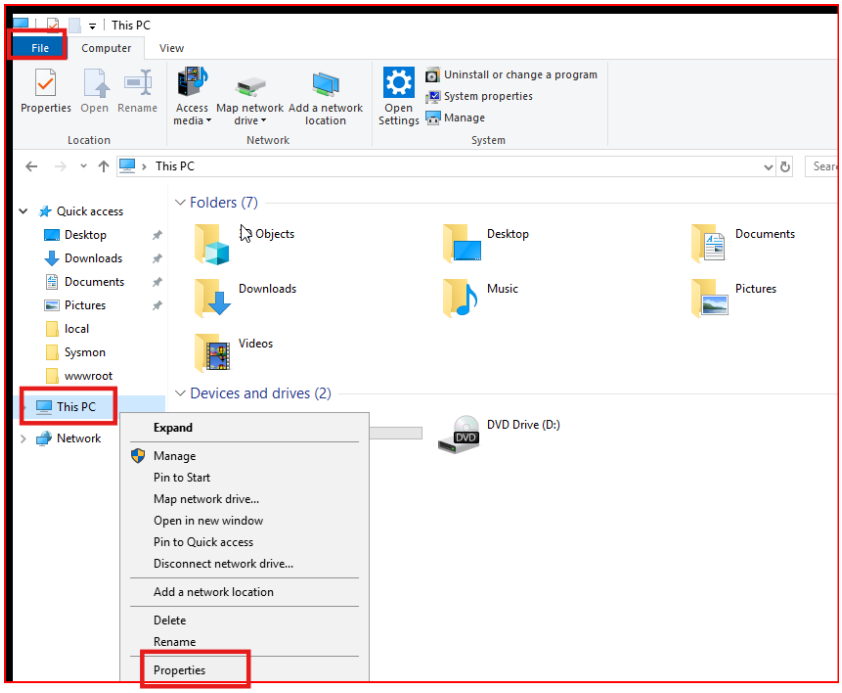
Repeat **the same steps** as Phase 1 on the **Backup Server**.



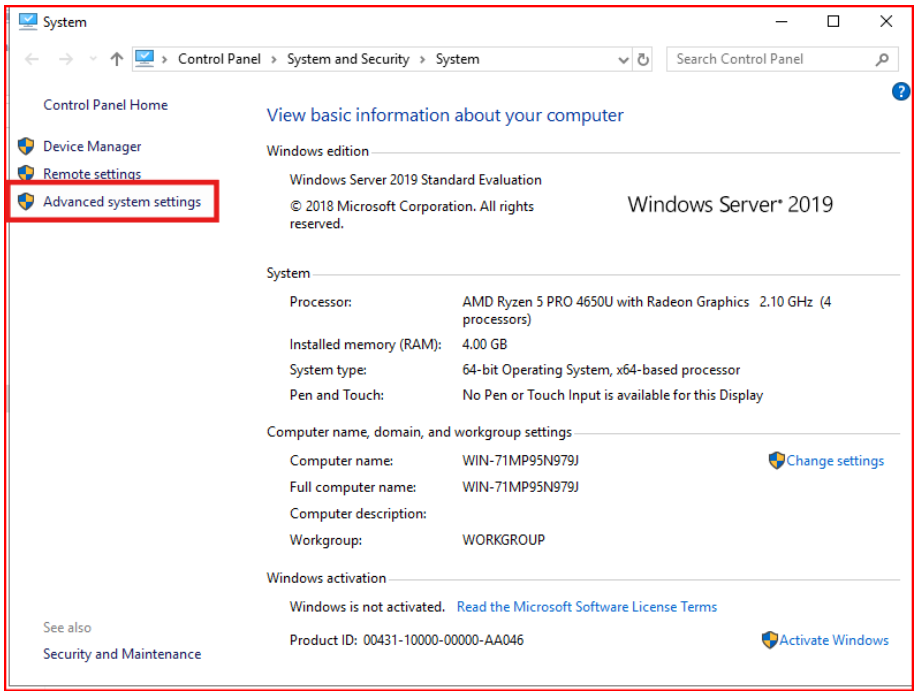
Set Primary DNS Suffix (Recommended – Both Servers)

Do this on **both** Web Server and Backup Server:

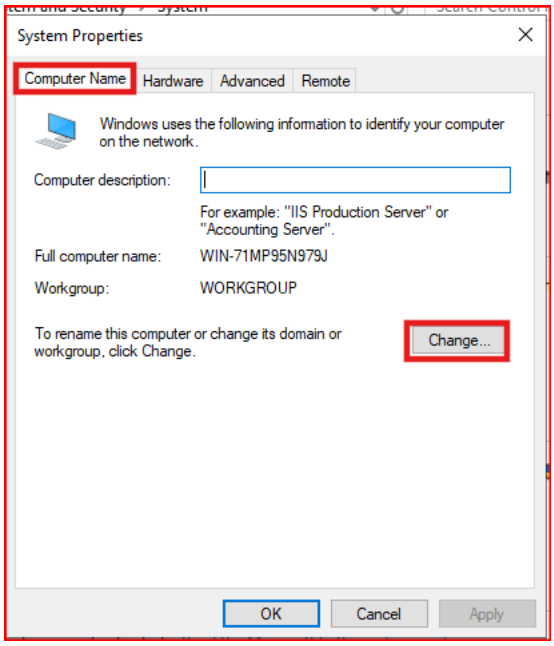
1. Open **File Explorer** → right-click **This PC** → **Properties**.



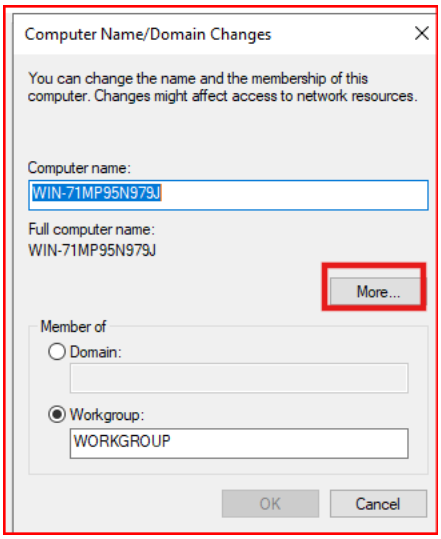
2. Click **Advanced system settings**.



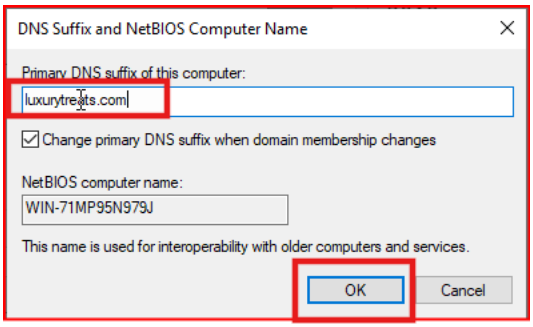
3. Click the **Computer Name** tab → click **Change…**.



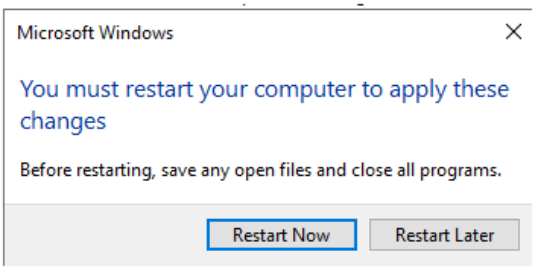
4. Click **More...**.



5. In "Primary DNS suffix of this computer" type: luxurytreats.com
6. Click **OK** → **OK** → **OK**.

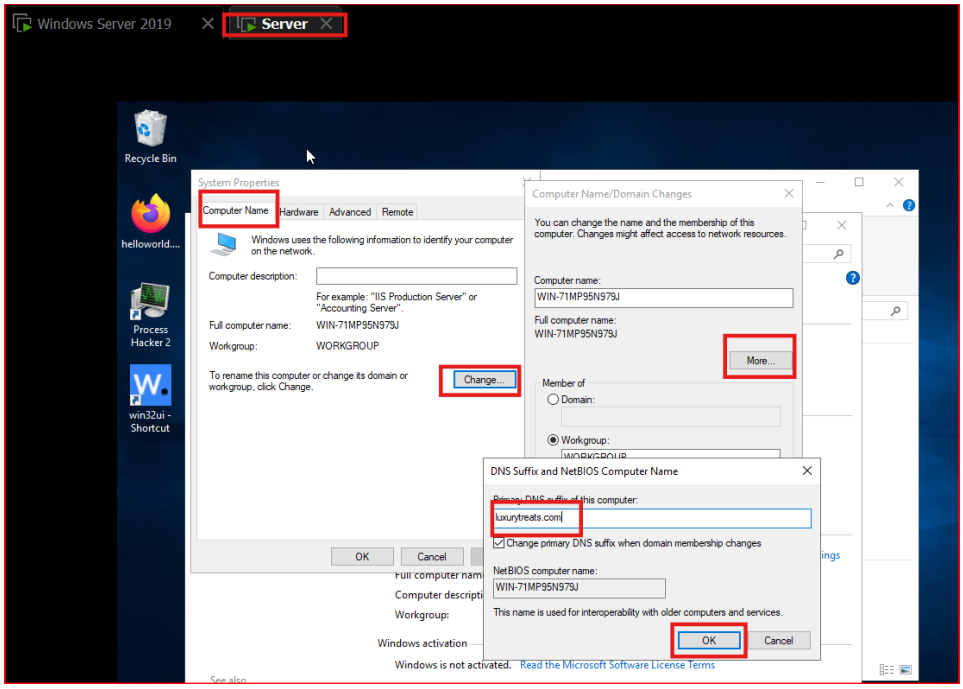


7. Click **Restart Now**.



Do this Same on Backup Server

Verification



After both server's restart, log back in.

Enable Required Rules (Even if Firewall is Off)

On **both servers**, run these commands in PowerShell (Admin):

```
Enable-NetFirewallRule -DisplayGroup "Network Load Balancing"

Enable-NetFirewallRule -DisplayGroup "Windows Management Instrumentation (WMI)"

Enable-NetFirewallRule -DisplayGroup "Remote Service Management"

Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing"
```

```
PS C:\Users\Administrator> Enable-NetFirewallRule -DisplayGroup "Network Load Balancing"
PS C:\Users\Administrator> Enable-NetFirewallRule -DisplayGroup "Windows Management Instrumentation
 (WMI)"
PS C:\Users\Administrator> Enable-NetFirewallRule -DisplayGroup "Remote Service Management"
PS C:\Users\Administrator> Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing"
PS C:\Users\Administrator>
```

Restart NLB Related Services

On **both servers** run:

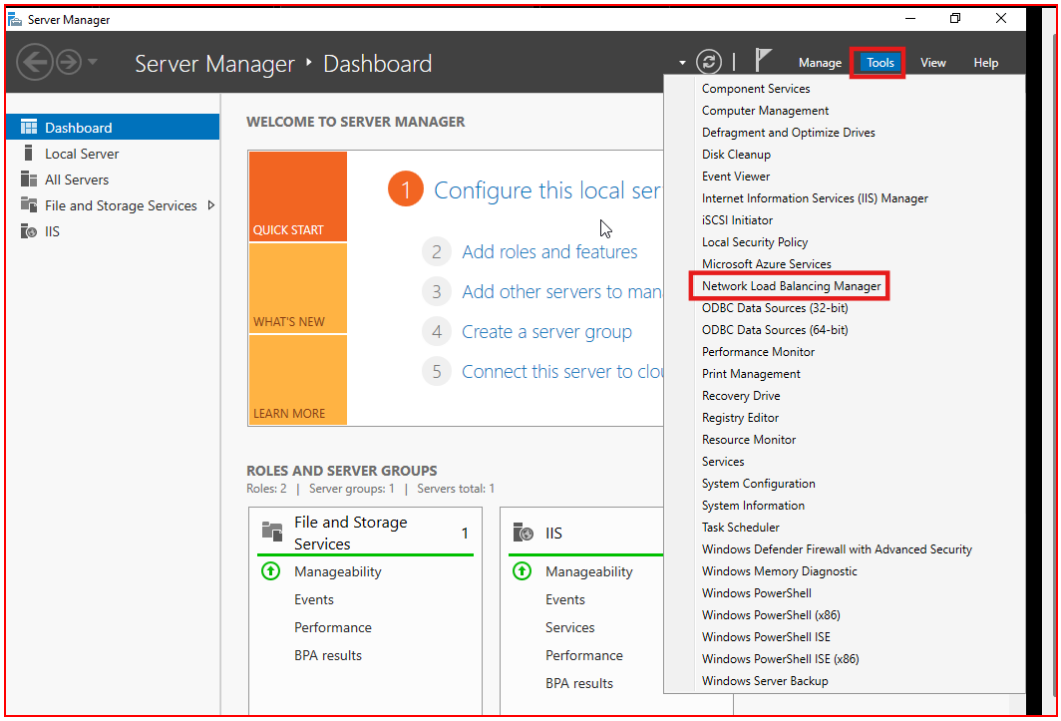
```
Restart-Service wlbs -Force

Restart-Service winmgmt -Force
```

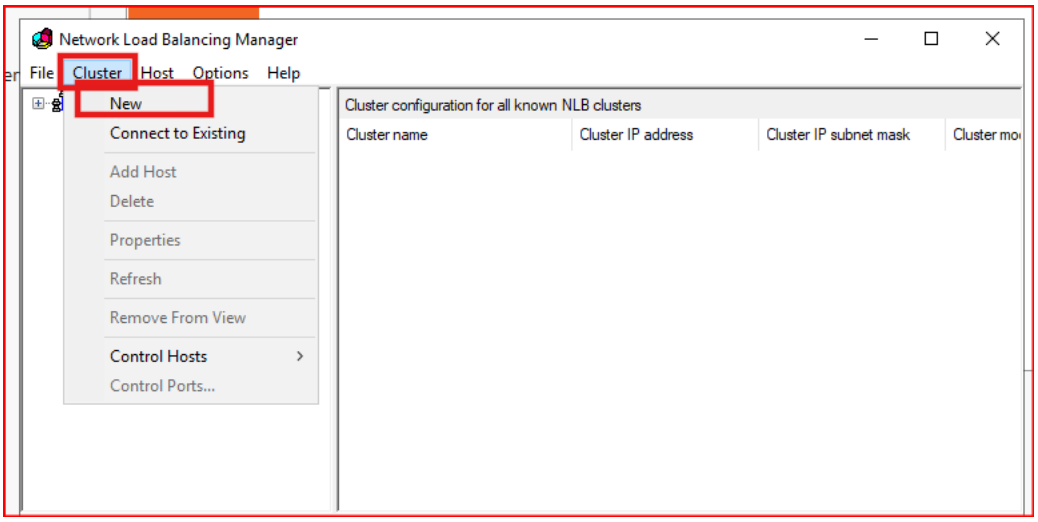
```
PS C:\Users\Administrator> Restart-Service wlbs -Force
PS C:\Users\Administrator> Restart-Service winmgmt -Force
WARNING: Waiting for service 'User Access Logging Service (UALSVC)' to start...
WARNING: Waiting for service 'User Access Logging Service (UALSVC)' to start...
PS C:\Users\Administrator>
```

Create the NLB Cluster (on Web Server)

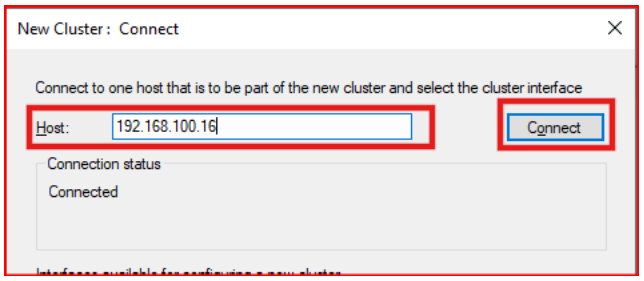
1. On **Web Server**, open **Server Manager → Tools → Network Load Balancing Manager**.



2. Right-click **Network Load Balancing Clusters → New Cluster**.

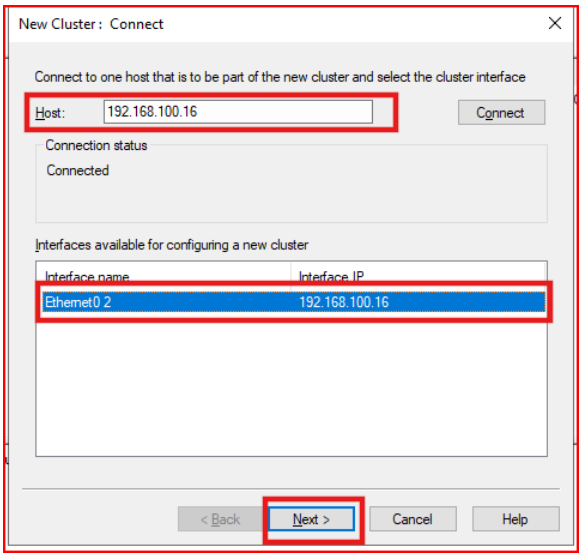


3. In the Host field type: 192.168.100.16 → click **Connect**.

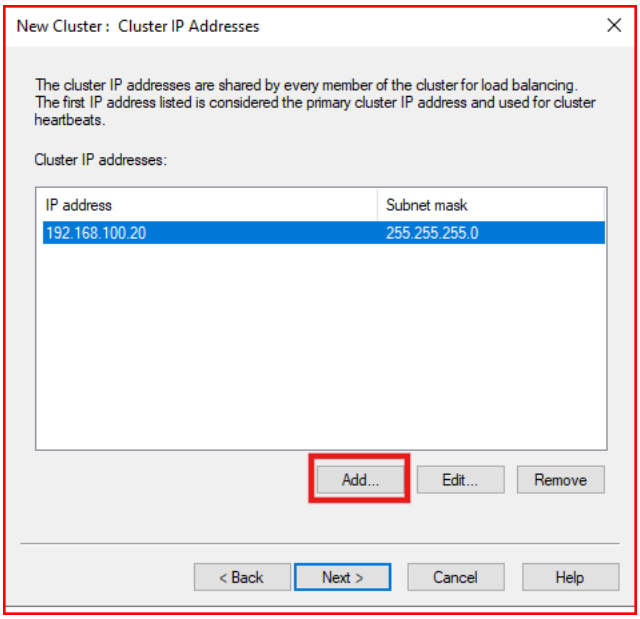


4. Select the interface that shows IP 192.168.100.16 → click **Next**.

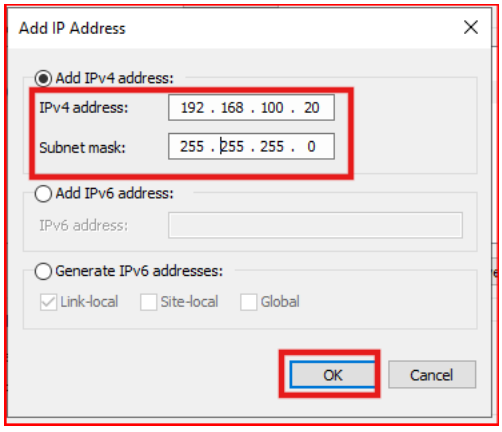
5. On the next screen click **Next** (leave Priority as 1).



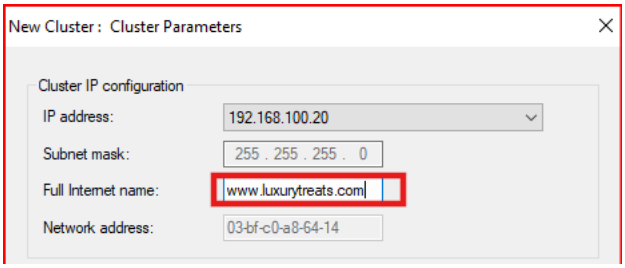
6. Click **Add** to create a Cluster IP address:

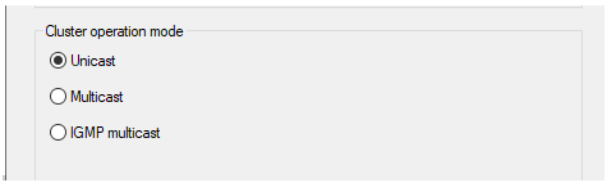


- IP address: 192.168.100.20 (this is the Virtual IP)
- Subnet mask: 255.255.255.0
- Click **OK** → **Next**.

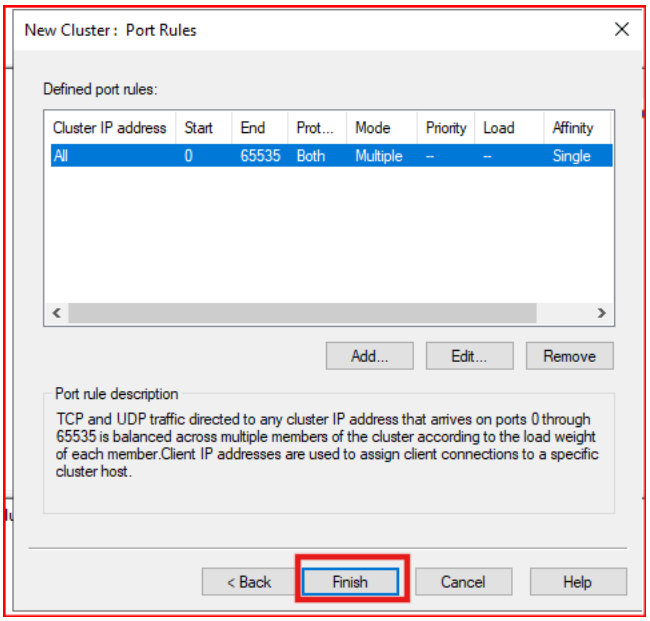


7. Full Internet name: www.luxurytreats.com
8. Cluster operation mode: **Unicast** (recommended) → **Next**.

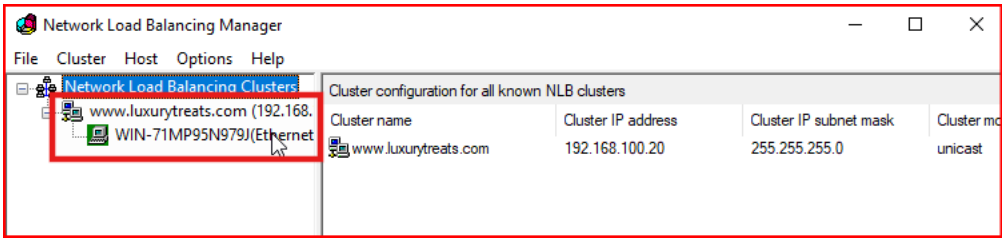




9. Leave port rules as default → click **Finish**.

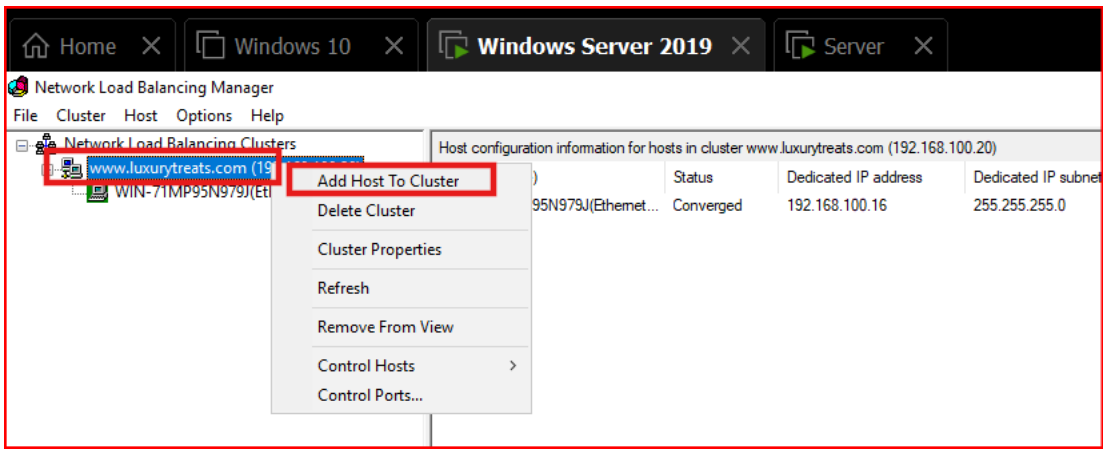


The first node should now show status **Converged**.



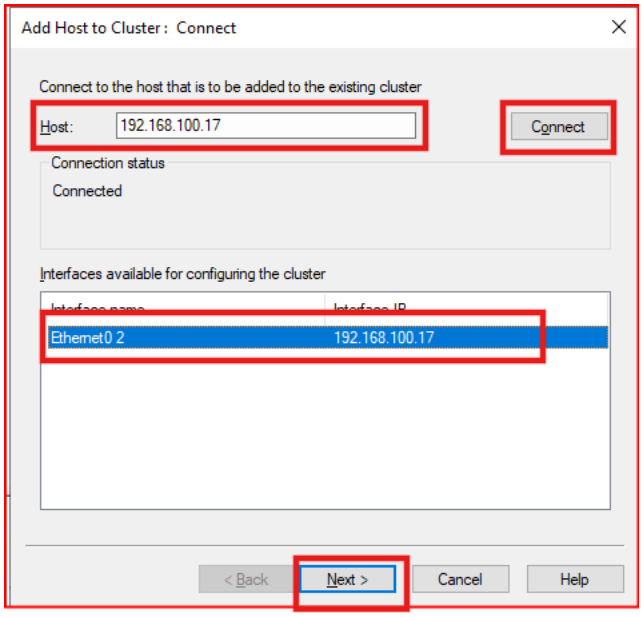
Add Backup Server to the Cluster

1. In Network Load Balancing Manager, right-click the cluster you just created → **Add Host to Cluster**.

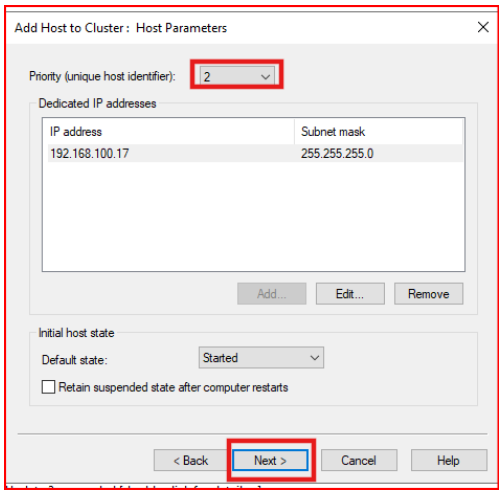


2. Type Host: 192.168.100.17 → click **Connect**.

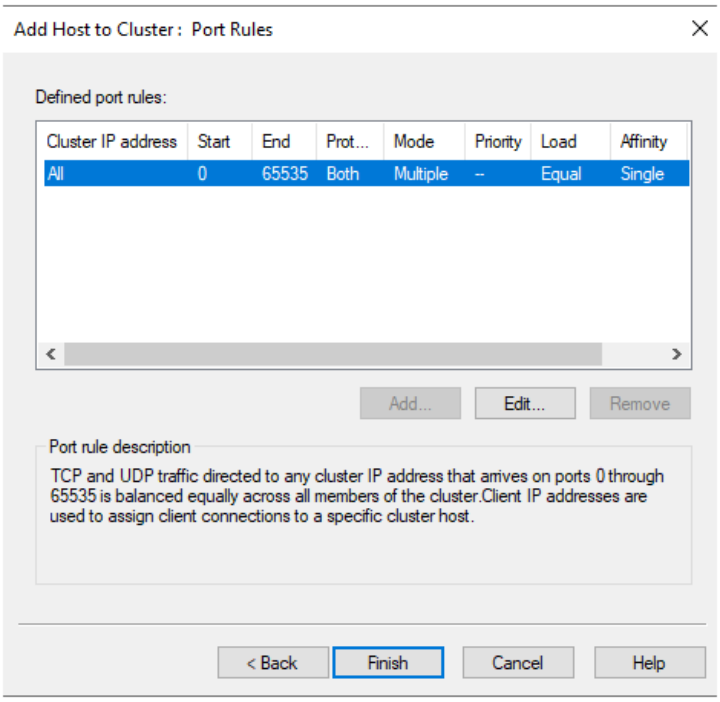
3. Select the interface with IP 192.168.100.17 → click **Next**.



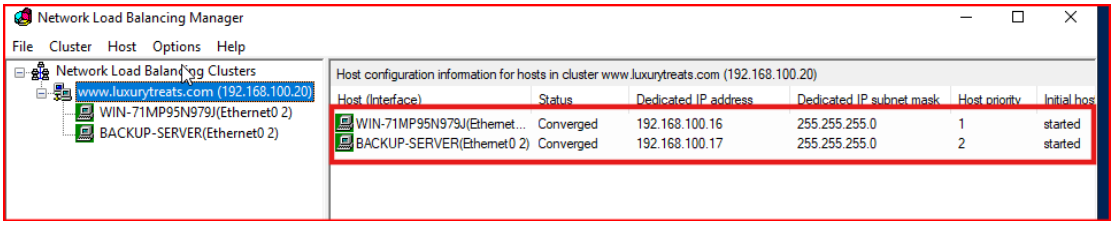
4. Leave Priority as 2 → click **Next**.



5. Click **Finish**.



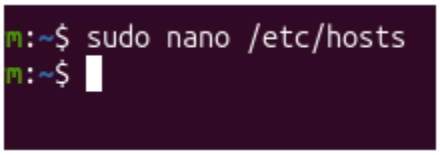
Both nodes should now show status **Converged**.



Configure Name Resolution on Attacker Machine

1. Log in to **Attacker Machine**.
2. Open terminal and run:

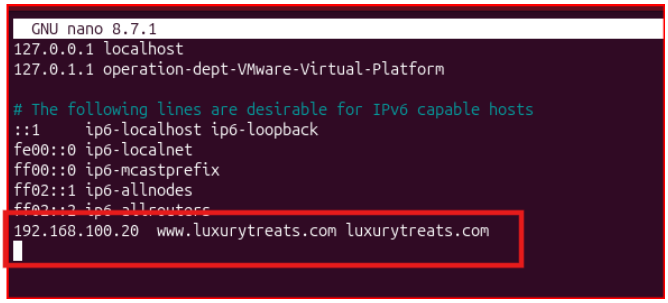
```
sudo nano /etc/hosts
```



3. Add this line at the bottom:

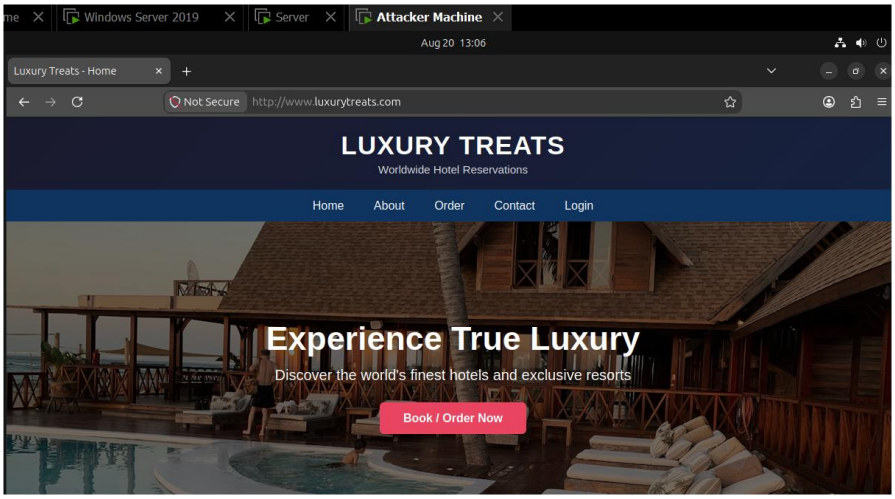
```
192.168.100.20     www.luxurytreats.com
```

4. Save and exit (Ctrl+O → Enter → Ctrl+X).



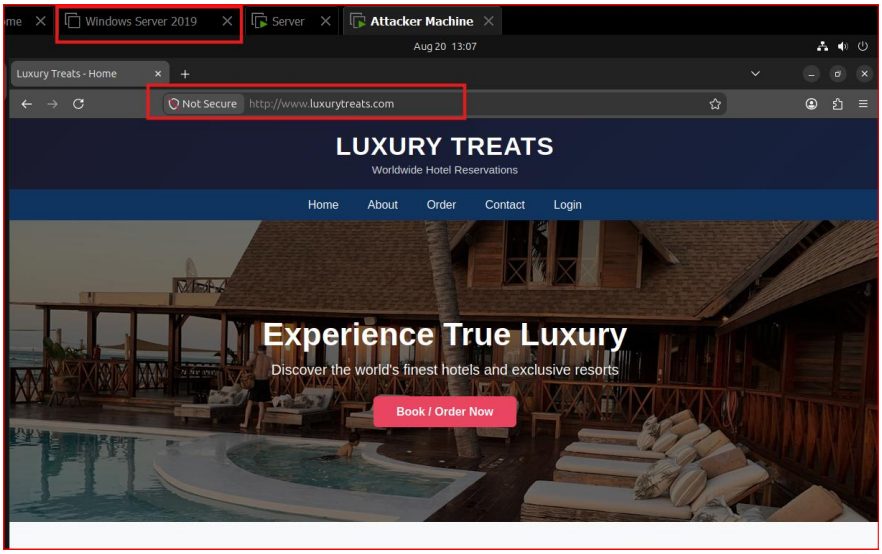
Test the Cluster (Before Failover)

1. On Attacker Machine open Firefox.
2. Go to: http://www.luxurytreats.com
3. Confirm the website loads correctly.



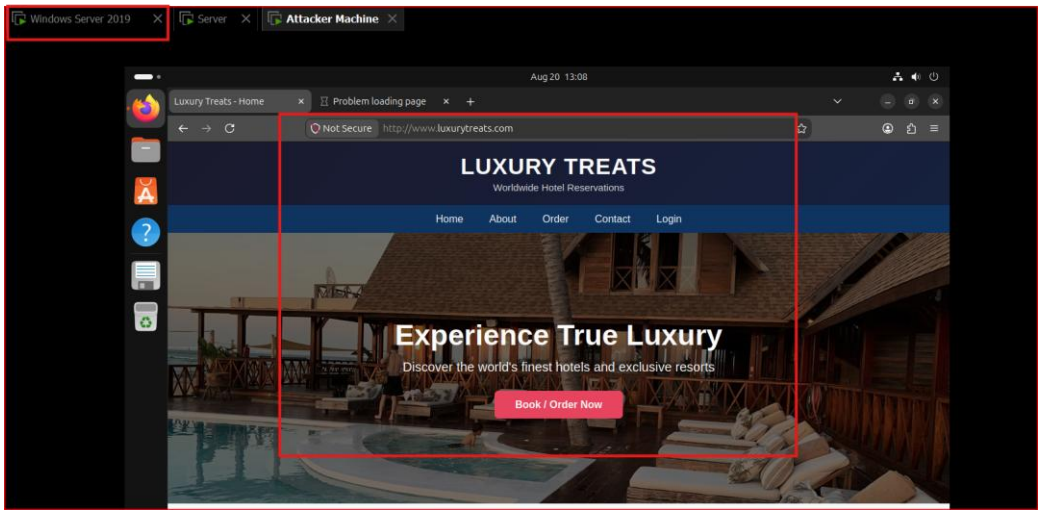
Failover Test (Most Important)

1. On **Web Server**, shut it down completely (Start → Power → Shut down).
2. Wait 10–20 seconds.
3. On Attacker Machine, refresh the browser (http://www.luxurytreats.com).
4. The website should still load (now served by Backup Server).



Recovery Test

1. Power on the **Web Server** again.
2. Wait until it fully boots and joins the cluster (check in NLB Manager – both nodes should become Converged again).
3. Refresh the website on Attacker Machine – it continues working.



Conclusion

Successfully implemented a two-node Network Load Balancing cluster for high availability.

Key Results:

- Both servers share the Virtual IP 192.168.100.20
- Clients access the site using one name (www.luxurytreats.com)
- When Web Server fails, Backup Server automatically takes over
- When Web Server recovers, it rejoins the cluster automatically

This is a real-world technique used for business continuity and disaster recovery of critical web services.